

MANPREET SINGH

Senior React Native AI Developer | AI Integrations | Expo | React JS | Voice & LLM Systems

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EXECUTIVE SUMMARY

Senior Mobile Engineer with 9+ years of experience building, scaling, and modernizing high-impact mobile and tvOS applications across fintech, banking, e-commerce, and healthcare domains. Strong expertise in React Native architecture, performance optimization, and delivering production-ready applications used by millions of users.

Experienced in integrating AI/LLM systems into mobile applications including voice-to-LLM pipelines, real-time audio streaming, and AI-driven chat experiences. Skilled in working with OpenAI, Claude, and other LLM APIs with focus on latency optimization, guardrails, and production-grade logging.

CORE SKILLS

Mobile & Web Platforms

React Native, Expo, Android, iOS, React.js, tvOS

Languages

TypeScript, JavaScript, python(Basic)

State & Data Management

Redux, TanStack Query, Context API, REST APIs

Native, Real-Time & Media

Custom Native Modules, WebRTC, AudioRTC, Bluetooth Audio, Real-time PCM Streaming, Third-party SDK Integrations

Architecture & Quality

Modular Architecture, Scalable Component Design, Performance Optimization, Unit Testing (Jest)

AI & LLM Integration

LLM API Integration (OpenAI, Claude)
Voice-to-LLM pipelines (Speech → Text → LLM → Audio)
Real-time streaming responses (WebSockets / SSE)
Prompt engineering & response structuring
AI guardrails, moderation & safety handling
Token optimization & cost control strategies
Context management & memory systems
Text-to-Speech (TTS) & Speech-to-Text (STT)

CI/CD & Release

GitHub Actions, TestFlight, Play Store, Firebase App Distribution, CodePush, EAS Builds

Backend & Services

Firebase, fastAPI (Basic knowledge)

Tools

GitHub, Bitbucket, GitLab, Copilot, Claude Sonnet, Claude Opus, Cursor, OpenAI API

PROFESSIONAL EXPERIENCE

Senior React Native Developer

Freelancing | Sept 2025 – Present

- Building scalable mobile applications with AI-powered features
- Integrating LLM APIs and voice-based interaction systems
- Delivering end-to-end mobile solutions with performance optimization

Senior Consultant

Xebia India, Gurgaon | Sept 2021 – Sept 2025

- Led development of large-scale React Native applications in fintech and banking
- Designed scalable architectures and reusable component systems
- Improved app performance, stability, and release processes

React Native Developer

Freelancing | April 2019 – Sept 2021

- Delivered multiple production-grade mobile applications across domains

- Integrated third-party SDKs including payments, analytics, and media

React Native Developer

Mobilyte Solutions, Mohali | June 2018 – April 2019

- Built and maintained mobile applications with focus on UI/UX and performance

Junior React Native Developer

Igniva Infotech, Mohali | Mar 2017 – June 2018

- Contributed to mobile app development and bug fixing

SELECTED TECHNICAL ACHIEVEMENTS

- Built real-time voice-to-LLM pipeline with interruption handling (GPT-style streaming)
- Implemented low-latency AI conversational systems using WebRTC & PCM streaming
- Integrated LLM-based chat systems with streaming responses and context handling
- Designed production-grade logging & monitoring for AI pipelines
- Developed AI-powered audio interaction systems with TTS/STT integration
- Built high-performance custom native audio libraries supporting multiple streams
- Strong expertise in Bluetooth audio routing, reconnection, and device handling

KEY PROJECTS

IDFC First Bank — Senior React Native Engineer

- Designed and delivered secure mobile banking features including payments, transfers, and account management
- Built dispute and fraud reporting flows
- Automated backend issue creation via APIs, reducing support workload
- Ensured compliance with banking security standards
- Explored AI-driven automation for intelligent support workflows

Tonik Bank — Senior React Native Engineer

- Upgraded React Native versions across the application
- Refactored native modules for performance and security improvements
- Resolved complex production issues and crashes

Ferns N Petals UAE — React Native Engineer

- Built e-commerce application with multi-currency and region-specific flows
- Integrated payment gateways and logistics services

HelloBible — Senior React Native Engineer

- Implemented AI-powered conversational chat using LLM APIs
- Built text-to-speech pipeline for natural AI responses
- Designed prompt structures and context handling
- Optimized latency for real-time AI interactions

TopMusic — React Native Engineer

- Improved app stability by fixing critical production issues
- Optimized rendering and data-fetching performance

KEY ACHIEVEMENTS & RESPONSIBILITIES

- Delivered mobile applications used by millions of users globally
- Reduced operational costs through workflow automation
- Modernized legacy React Native codebases without disruption
- Mentored developers and conducted code reviews
- Reduced technical debt through better architecture decisions
- Implemented analytics and crash monitoring (Firebase Crashlytics, Adobe Analytics)

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering

Chitkara University, Himachal Pradesh | 2011 – 2015